



University of Asia Pacific

Department of Computer Science and Engineering

Course Title: Robotics Lab

Course Code: CSE 438

Annexure Component (CEP Mapping)

Submitted To

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CEP Mapping

1. Knowledge Profile (K's) Mapping:

Knowledge Attribute	Description	Evidence in Report	Section/Subsection
WK3	Engineering fundamentals required in the discipline	Design and integration of sensors, motors, pump, controllers, and communication modules	II-A. System Architecture, II-C. Hardware Components, II-E. Working Principle
WK4	Specialist engineering knowledge	Application of IoT, ESP32, ESP32-CAM, WebSocket communication, mDNS, and autonomous robotics	II-A – II. D System Design and Methodology
WK6	Knowledge of engineering practice and technology	Practical implementation using Arduino IDE, ESP32 libraries, web dashboard, and hardware testing	III-A. System Setup, III-B. Testing and Results
WK7	Role of engineering in society	Fire safety solution aimed at reducing risk to human life and property	V-B. Social Relevance
WK8	Research literature and emerging technologies	Study and comparison of existing firefighting robots and IoT-based safety systems	I-D. Related Work

2. Complex Engineering Problems (P's) Mapping

Problem Attribute	Description	Evidence in Report	Section/Subsection
WP1	Requires in-depth engineering knowledge	Integration of embedded systems, robotics, wireless communication, and sensor technologies	II. System Design and Methodology
WP3	Requires analysis and formulation of solutions	Development of fire-detection logic and autonomous navigation algorithm based on sensor inputs	II-E. Working Principle and Data Flow
WP7	Involves interconnected subsystems requiring a systems approach	Coordination among sensors, motors, pump, ESP32 modules, camera system, and web dashboard	II-A. System Architecture

3. Complex Engineering Activities (A's) Mapping

Activity Attribute	Description	Evidence in Report	Section/Subsection
EA1	Use of diverse resources and technologies	Utilization of ESP32, ESP32-CAM, sensors, actuators, software libraries, and networking technologies	II-C. Hardware Components, II.D Software Components

EA2	Resolution of interactions among technical components	Integration of sensing, control, communication, monitoring, and user interface functionalities	II-A. System Architecture, II.B Communication Flow
EA3	Innovation and research-based solution development	Hybrid autonomous/manual firefighting robot with live video streaming and IoT dashboard	II-A. System Architecture, II-E. Working Principle and Data Flow, IV-A. Functional Outcomes
EA5	Application of principles-based approaches beyond previous experiences	Development, testing, and refinement of a complete prototype system	III. Implementation and Results, V. Future Work